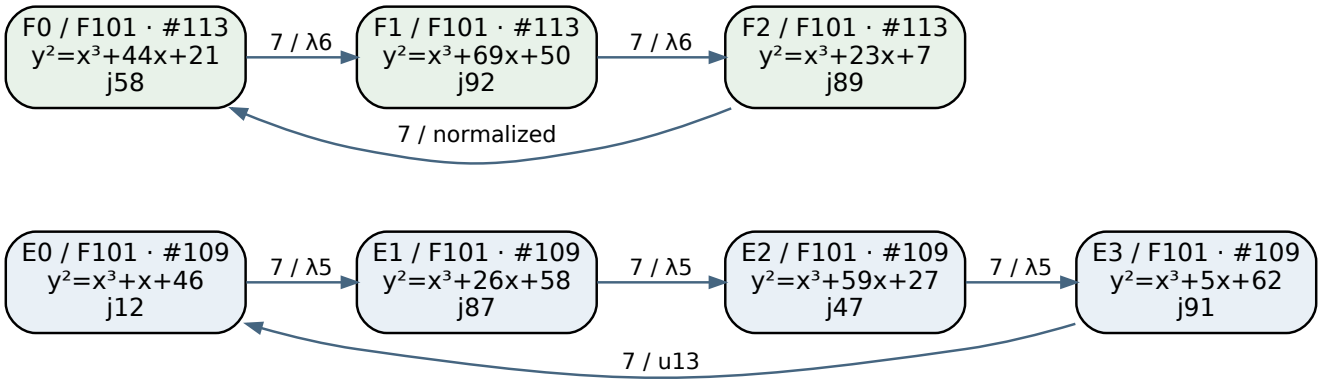
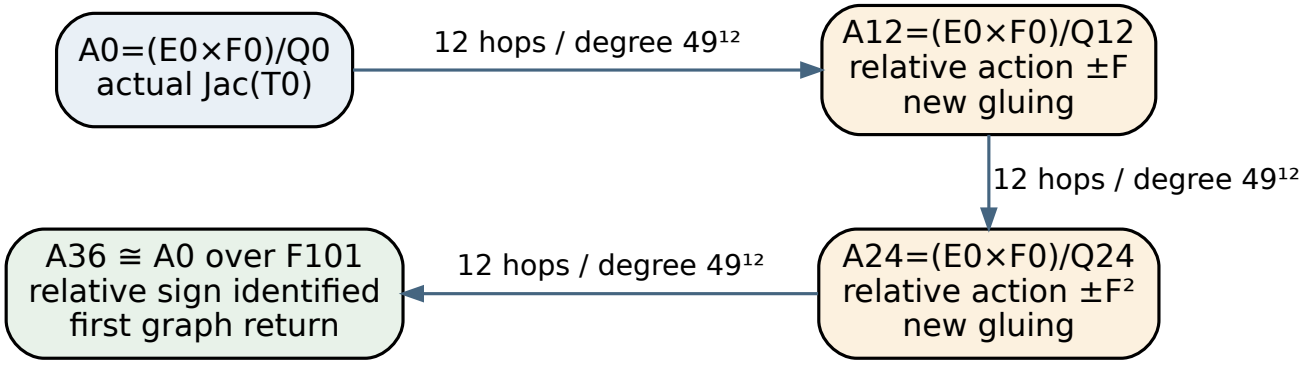
Actual gluing return36 over F101

2026-11-12 · complete native33 · written catalogf48205 · unsigned

The full changed suite passes33 tests with zero failures,ignores or filters; oldguards,formatting,strictClippy and all621 source/fixture hashes pass. Native70a16a retains whole75,828-byte output as standardXZ9,336,source/status/ hashes/exact roundtrip. Targeted1/32filtered367c83 is separate. [pcc:unverified]



Actual short-Weierstrass factor models and normalized degree7 maps. E/F orders109/113,traces-7/-11,selected eigenlines5/6. Native exhausts every generic j/both twists and all exceptional models:exactly4+3 factor classes. [pcc:unverified]



Each paired hop has degree49. E0/F0 are defined above;the quotient kernel Q has16 actual4-torsion points. The elliptic pair returns12,while the actual 4-graph and its2-torsion reduction first return36. All36 states are compared pairwise;first two transported graphs match separately extractedT1/T2. [pcc:unverified]

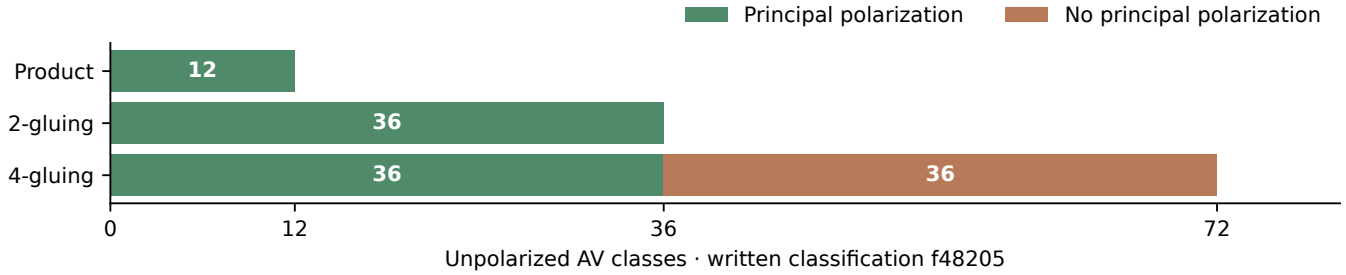
At every step the checker verifies all16 graph points,Frobenius stability, all256 sums and inverse Weil pairs. Across12 factor pairs it finds12 Galois 4-isomorphisms each:six anti-isometries/three relative-sign orbits and three other sign orbits. The walk exhausts all36 anti4 and36 two-graph classes. Each factor has15 additive subgroups of its16-point4-group;exactly three are Frobenius-stable,of sizes1,4,16. Subsets omitting identity are pre-excluded. [pcc:unverified]

Global identification and prime-field transfers

All factor coefficients,scales and kernels are defined overF101. Standard quotients by the stable graphs and the universal property descend each map. Degree49 is coprime to109/113,so both prime DLP subgroups are preserved. The genus-two existence/CM/quotient/degree/polarization bridges are written in ISOGENY_CYCLE_PROOF_V2.md/f48205. Other targets have quotient presentations; no new full Mumford maps or72 sextic equations were evaluated. [pcc:unverified]

The CM endomorphisms of the short factor cycles are-(31+5pi) and22+pi. Modulo4,their relative12-hop action is Frobenius up to sign,of exact order3 modulo signs. Hence neither12 nor24 closes the actual graph. Complete native coordinates and kernel-checked finite monodromy corroborate the written ordinary-CM ideal argument;finite point agreement alone is not a global map identity. [pcc:unverified]

The fixed quotient catalog and rational control



Written classification valuesf48205,not measurements or native enumeration of whole AVs. Complete native factor/graph inputs70a16a support the stated standard global bridges. [pcc:unverified]

The canonical intrinsic-factor sum has kernel killed by4,using the real Weil gluing exponent. Stable subgroup types force product,two-glue or four-glue. Unique elliptic factors and their±1 automorphisms give 12(1+3+6)=120 unpolarized classes. A principal pullback must have equal diagonal degrees n,n:ab=n² plus both kernel-length bounds forces this. Thus only anti graphs qualify:84 principal=12products+36two+36four; 36 non-anti4 classes admit no principal polarization. The72 nonproduct principal quotients are genus-two Jacobians overF101 by primary gluing classification;Torelli gives uniqueness. These are written global results, not72 newly tested curve models. [pcc:unverified]

Closed36 rational action

Native checks both short-cycle CM signs on all16 geometric4-points and all 109/113 rational points. After36 hops the residues are45/56;the final relative F-factor sign gives45/57. PublicCRT12035 matches the actual outer K0·factor-walk·H0/4 route on every12,317 oldT0 class;scalar12036 differs on all 12,316 nonidentity classes. Globally the endomorphism has one-sided CM prime support,not integer multiplication. The huge7-primary kernel is written, not enumerated. No timing protocol or cost improvement is inferred. [pcc:unverified]

Formal checks,certificates and failures

Nine bound local-pass IsogenyCycles declarations cover universal norm and two cycle polynomials,exact norms,all16-vector monodromy,relative-sign nonclosure,surjective graph obstruction,CRT and forced principal diagonals. Gate03 passes9/zero trusted native;project census169 written/33 modules. Actual varieties and standard global bridges remain outside the formal model. [pcc:unverified]

Algebra02/8d3d02 passes23 tests and10 source guards;the two old certificates are byte-identical. New IDC1hcf1c09671f79d47d,dc15986d66227afa,93a8cf047d0ece53 each use32 issuer+64 extra checker points over2^61-1 and exact Lean03 digest. These evaluated records and local Lean results supply no independent scientific signature. Staticstylecf3e19,formal product-quantifier586ae2 and algebra syntactic-degree5f671c failures remain immutable;all criteria retained. [pcc:unverified]

Sources:Galbraith25/26;Bruin–Doerksen0902.3480 Theorem1.1/Lemma2.4/2.6/ Theorem3.2;Howe–Lauter1202.6308 §2;Milne I7/8/13,IV2/4. Characteristic101 conditions checked;characteristic0 Proposition2.8 not used. ExternalCT43/44 cc5d5541 abstract work credited,original blobs retained. KB/ atlas inaccessible; no novelty inference. REPORT.md retains full source links,reproduction, requirement table and scoped limitations. [pcc:unverified]

Next:actual unbalanced(7,1) two-glue and(49,1) four-glue constructions with whole rational/exception controls. Consumer1509/8f456314 stays fixed pending ownCI;root0796 check14:37,consumer14:47. Merge ledger/C/signing and live-budget/ S3/vault blockers persist;no paid experiment or benchmark. [pcc:unverified]